

2(a)	sum / total momentum before (interaction) = sum / total momentum after or sum / total momentum (of a system of interacting objects) is constant	[1]
	if no resultant force / for an isolated system	[1]

(b)(i)	$3m \times 4 = m \times v \sin \theta$ $(v \sin \theta = 12)$ $2m \times 6 = m \times v \cos \theta$ $(v \cos \theta = 12)$ therefore, $\sin \theta = \cos \theta$ or $\tan \theta = 1$ $\theta = 45^\circ$ or a method using close triangle of the momentum vectors and trigo	[1] [1] [1]
(b)(ii)	$mv \times \cos 45^\circ = 12m$ or $mv \times \sin 45^\circ = 12m$ or $(mv)^2 = (3m \times 4)^2 + (2m \times 6)^2$ $v = 17 \text{ m s}^{-1}$ or use pyth theorem for momentum	[1-ecf] [1-ecf]
(c)	Chemical energy = $\frac{1}{2} mv^2$ (By conservation of energy) Chemical energy = 0.0050×700 or $5.0 \times 0.700 = 3.5$ $3.5 = (0.5 \times 3m \times 4^2) + (0.5 \times 2m \times 6^2) + (0.5 \times m \times 17^2)$ 0.0050×700 or $5.0 \times 0.700 = 3.5 = 204m$ $m = 0.017 \text{ kg}$	[1] [1-ecf] [1-ecf]

2(a)	$E = q\phi$	
------	-------------	--